

Algebra II with Trigonometry

Chapter 6.4

Study guide

Saved tutoring response with MathJax rendering. Original PDF file:
[4533650_alg-2trig-h-chp-6-4-note.pdf](#)

Certainly! Here is a self-contained study guide and summary based on the provided PDF.

1. Short Summary

This document covers the graphs and periodic properties of four key trigonometric functions: **tangent (tan)**, **cotangent (cot)**, **secant (sec)**, and **cosecant (csc)**. It explains their periods, describes how to find vertical asymptotes, and provides intervals for graphing both basic and transformed (shifted/scaled) versions of these functions. Examples and key formulas for graphing are included.

2. Core Definitions, Theorems, and Formulas

Definitions & Properties

- **Tangent (tan x):**
 - Period: π
 - Vertical asymptotes: $x = \frac{\pi}{2} + n\pi, n \in \mathbb{Z}$
- **Cotangent (cot x):**
 - Period: π
 - Vertical asymptotes: $x = n\pi, n \in \mathbb{Z}$
- **Cosecant (csc x):**
 - Period: 2π

- Vertical asymptotes: $x = n\pi, n \in \mathbb{Z}$
- Defined by: $\csc x = \frac{1}{\sin x}$
- **Secant (sec x):**
- Period: 2π
- Vertical asymptotes: $x = \frac{\pi}{2} + n\pi, n \in \mathbb{Z}$
- Defined by: $\sec x = \frac{1}{\cos x}$

Important Periodicity Properties

- $\tan(x + \pi) = \tan x$
- $\cot(x + \pi) = \cot x$
- $\csc(x + 2\pi) = \csc x$
- $\sec(x + 2\pi) = \sec x$

Intervals for Graphing One Period

Basic Functions

- $y = a \tan(kx)$: $(-\frac{\pi}{2k}, \frac{\pi}{2k})$
- $y = a \cot(kx)$: $(0, \frac{\pi}{k})$
- $y = a \csc(kx)$ or $y = a \sec(kx)$: $[0, \frac{2\pi}{k}]$

Shifted/Transformed Functions

- $y = a \tan k(x - b)$: $(-\frac{\pi}{2k} + b, \frac{\pi}{2k} + b)$
- $y = a \cot k(x - b)$: $(b, \frac{\pi}{k} + b)$
- $y = a \csc k(x - b)$ or $y = a \sec k(x - b)$: $[b, \frac{2\pi}{k} + b]$

3. Step-by-Step Study Guide

A. Graphing Tangent and Cotangent

1. Tangent: $y = a \tan(kx + c)$

- **Period:** $\frac{\pi}{k}$
- **Asymptotes:** Solve $kx + c = \frac{\pi}{2} + n\pi$
- **To Graph One Period:**
- Start: $x = -\frac{\pi}{2k}$ (for standard kx)

- End: $x = \frac{\pi}{2k}$

2. Cotangent: $y = a \cot(kx + c)$

- **Period:** $\frac{\pi}{k}$
- **Asymptotes:** Solve $kx + c = n\pi$
- **To Graph One Period:**
- Start: $x = 0$
- End: $x = \frac{\pi}{k}$

B. Graphing Secant and Cosecant

1. Cosecant $y = a \csc(kx + c)$

- **Period:** $\frac{2\pi}{k}$
- **Asymptotes:** Where $\sin(kx + c) = 0$ (i.e., $kx + c = n\pi$)
- **To Graph One Period:**
- Interval: $[0, \frac{2\pi}{k}]$
- **Tip:** Graph $y = a \sin(kx + c)$ first, then draw vertical asymptotes where it crosses zero, and sketch branches for csc.

2. Secant $y = a \sec(kx + c)$

- **Period:** $\frac{2\pi}{k}$
- **Asymptotes:** Where $\cos(kx + c) = 0$ (i.e., $kx + c = \frac{\pi}{2} + n\pi$)
- **To Graph One Period:**
- Interval: $[0, \frac{2\pi}{k}]$
- **Tip:** Graph $y = a \cos(kx + c)$ first, then draw vertical asymptotes where it crosses zero, and sketch branches for sec.

C. General Steps for Graphing Any Transformation

1. **Find the period:** Use $\frac{\pi}{k}$ for tan/cot, $\frac{2\pi}{k}$ for sec/csc. 2. **Find the interval:** Use formulas above for shifts. 3. **Locate asymptotes:** Set the denominator of the trig fraction to 0. 4. **Draw guide graph:** Plot basic sine/cosine where needed. 5. **Draw the function:** Add asymptotes and sketch increasing/decreasing behavior.

4. Practice Problems with Solutions

1. What is the period of $y = 3 \tan 2x$?

Answer: $\frac{\pi}{2}$ (since period = $\frac{\pi}{k}$ and $k = 2$)

2. Find the vertical asymptotes of $y = \cot(3x)$.

Answer: Set $3x = n\pi \implies x = \frac{n\pi}{3}$ for integer n .

3. State the interval to graph one period of $y = 2 \sec(4x)$.

Answer: $[0, \frac{2\pi}{4}] = [0, \frac{\pi}{2}]$

4. What are the vertical asymptotes of $y = \csc(x - \frac{\pi}{4})$?

Answer: Set $x - \frac{\pi}{4} = n\pi \implies x = n\pi + \frac{\pi}{4}$, $n \in \mathbb{Z}$.

5. Give the interval to graph one period of $y = 5 \cot(2x - \pi)$.

Answer: Use formula for cotangent with shift: interval is $(\frac{\pi}{2}, \frac{\pi}{2} + \frac{\pi}{2}) = (\frac{\pi}{2}, \pi)$, but using the document's format, $(0 + \frac{\pi}{2}, \frac{\pi}{2} + \frac{\pi}{2}) = (\frac{\pi}{2}, \pi)$.

5. Assumptions & Ambiguities

- The document uses color highlights and diagrams; text interpretation lacks the visual aids.
- Some parentheses in interval notation were incomplete in OCR; completed based on context.
- Implicitly, all periods and intervals assume $k > 0$.
- The direction to "graph one period" always means the domain interval for a complete cycle, not the range.
- For shifted functions, b denotes the horizontal (phase) shift.

If you need more help with any specific example problem or graph, just ask!